Theoretical Analysis of Blockchain-based Proportional Fair Scheduling for O-RAN

***Abstract*—Blockchain-based proportional fair scheduling (BC-PFS) provides a decentralized and trustworthy platform for cross-operator resource sharing and user scheduling in O-RAN. In the BC-PFS, trustworthy collaboration across operators largely relies on the blockchain consensus process, which may induce delayed scheduling and consensus failure. Yet the impact of blockchain consensus on the performance of BC-PFS in O-RAN remains unexplored. This paper establishes an analytical framework for BC-PFS and investigates the impact of delayed scheduling and consensus failure on system performance. We first develop a mathematical model of BC-PFS and prove the weak convergence of the throughput update process considering the delayed scheduling caused by relatively long block time. Moreover, we derive the semi-closed-form expression of user average throughput, along with its approximation and lower bound. We extend the model to evaluate throughput performance with both delayed scheduling and consensus failure. Furthermore, we reveal the trade-off between these two factors and derive the optimal scheduling interval. We also investigate the joint influence of channel coherence time and blockchain propagation delay on the optimal scheduling interval. The pooling effect is quantified by comparing the network utility of BC-PFS in O-RAN with that of conventional non-cooperative scheduling schemes. Finally, we conduct simulations to validate our proposed model and theoretical findings.**

***Index Terms*—Blockchain, O-RAN, proportional fairness, trustworthy, scheduling.**

I. Introduction

The open radio access network (O-RAN) has emerged as a transformative paradigm characterized by openness, programmability, intelligence, and disaggregated architecture with interoperable interfaces [1], [2]. Through network function virtualization and open interfaces, O-RAN shifts the traditionally closed RAN architecture toward a more open environment that allows broader participation. The demand for spectrum and RAN infrastructure sharing among multiple operators continues to grow in O-RAN [3]–[5]. This necessitates efficient and trustworthy resource coordination and user association across operators to improve resource utilization and guarantee service quality in O-RAN [6], [7]. However, the trust foundation for cross-operator user scheduling is often absent in O-RAN due to its openness and multi-stakeholder nature [8], [9]. Without trusted mechanisms, operators may have concerns about potential deceptions and become reluctant to share network resources and services with others [10]. These challenges highlight the pressing need for trustworthy scheduling paradigms that enable reliable collaboration in O-RAN with multiple operators that intend to cooperate.

To fill this gap, blockchain can be utilized as a critical enabler for establishing trust among multiple operators through immutable records and verifiable smart contract execution [11], [12]. Recent studies [13]–[15] have explored blockchain-enhanced inter-network collaboration to facilitate cross-operator coordination. Among them, the blockchain-based proportional fair scheduling (BC-PFS) is a representative approach for trustworthy cross-domain user scheduling. The BC-PFS provides an explicit trust mechanism required to enable reliable collaboration across heterogeneous networks in O-RAN. Specifically, smart contracts execute user scheduling strategies and update throughput records using immutable and verifiable code on-chain. User scheduling dynamically determines how radio resources are allocated among users according to the proportional fair rule [16], [17], where the user with the largest ratio of its instantaneous transmission rate to historical throughput is selected. The throughput is then updated based on the user selection and achieved user rate, and the updated throughput is used for subsequent scheduling decisions. The scheduling results are then recorded on a shared and immutable ledger through multi-party blockchain consensus. BC-PFS establishes inter-network trust for cross-operator scheduling by enabling verifiable scheduling execution and allowing operators to maintain a consistent and traceable view of the scheduling process in O-RAN [18], [19].

However, the trust establishment among multiple operators in O-RAN is not cost-free. Trustworthy coordination largely depends on the blockchain consensus process, through which participants reach agreement on shared network states and coordination decisions [20]–[22]. For BC-PFS in O-RAN, participating operators use blockchain consensus to jointly validate scheduling decisions and throughput updates. This process prevents any single operator from unilaterally modifying the shared scheduling state and ensures that all operators maintain a consistent, tamper-resistant, and verifiable record. The resulting shared state provides a trustworthy basis for cross-operator scheduling and enables transparent verification of BC-PFS execution. Although trust between network operators largely depends on the blockchain consensus process, its non-negligible impact on the performance of BC-PFS for O-RAN has been ignored.

The impact of blockchain consensus on BC-PFS mainly stems from the block time required for on-chain scheduling, i.e., the average time to create a new block that contains the latest scheduling result. On the one hand, the scheduling interval, or equivalently the block time, may be longer than the channel coherence time. In this case, the scheduling decision, made at the beginning of the scheduling interval, remains fixed throughout the scheduling interval and may become outdated as channel conditions vary. As a result, the mismatch between the scheduling decision and time-varying channel state degrades the BC-PFS performance. This is also known as delayed scheduling. Therefore, blockchains with a short block time are more conducive to mitigating delayed scheduling in

BC-PFS for O-RAN. Several blockchain platforms claim that they have reduced block times to the order of milliseconds [23], [24]. Nevertheless, even these relatively short block times may still be longer than the channel coherence time [25], [26]. This mismatch becomes more pronounced in high-mobility scenarios, where rapid fluctuations in channel state amplify the negative impact of delayed scheduling on the BC-PFS performance [27].

On the other hand, a short block time mitigates delayed scheduling but may leave insufficient time for consensus, thus increasing the risk of consensus failure for BC-PFS [28]. Specifically, the actual block propagation delay is variable, and blockchain maintainers may fail to complete consensus within the expected block time [29]. Such a failure further cancels a scheduling round and leads to no effective user throughput during that scheduling interval and degrades overall network performance of BC-PFS in O-RAN. Therefore, delayed scheduling and consensus failure exhibit a trade-off, and their impacts are intertwined, making the overall effect on BC-PFS performance intricate.

In a nutshell, BC-PFS enables trustworthy cross-operator user scheduling in O-RAN, but several critical challenges remain. First, the impact of the blockchain consensus process on BC-PFS performance remains insufficiently quantified. Second, how to balance the trade-off between delayed scheduling and consensus failure for BC-PFS in O-RAN is unexplored. Third, blockchain plays a dual role in BC-PFS. The negative impact of the consensus process may offset gains from blockchain-enhanced trustworthy resource pooling. Yet the conditions under which BC-PFS can achieve a positive pooling gain remain unclear.

In this paper, we establish an analytical model for BC-PFS by considering the impact of the consensus process. We reveal the trade-off between delayed scheduling and consensus failure and examine their combined impact on user average throughput. We provide theoretical guidance for selecting an appropriate scheduling interval for BC-PFS and derive the conditions under which a positive pooling gain can be achieved for O-RAN via BC-PFS. Through this, we provide a theoretical basis for the efficient use of BC-PFS in O-RAN. The main contributions of our paper are summarized as follows:

- We establish a mathematical model that captures the scheduling process across multiple operators for analyzing BC-PFS in O-RAN, and considers the impact of the blockchain consensus process.
- We characterize the impact of delayed scheduling on BC-PFS. We prove the weak convergence of the user throughput update process using stochastic approximation, and also derive the semi-closed-form expression for the average user throughput in O-RAN, along with its asymptotic behavior when the scheduling interval tends to infinity.
- We further analyze the impact of blockchain consensus failure and reveal the trade-off between it and delayed scheduling. We derive the optimal scheduling interval that balances these factors and examine how it varies with coherence time and blockchain propagation delay.

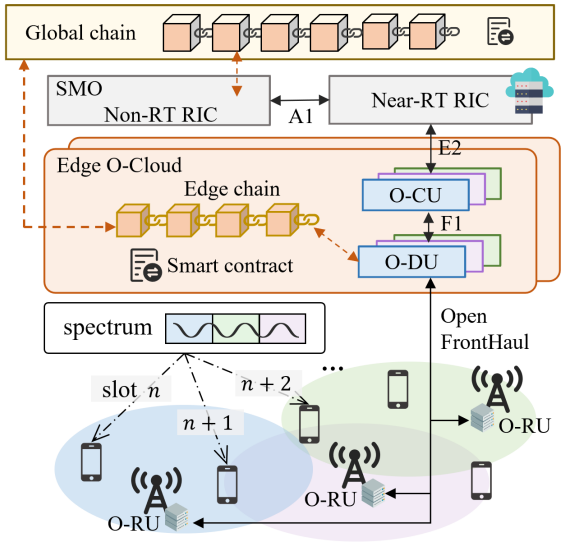


Figure 1. BC-PFS over multi-operator O-RAN.

- We examine the pooling effect introduced by BC-PFS in O-RAN. We derive necessary and sufficient conditions for achieving a positive pooling gain with delayed scheduling and consensus failure.
- We validate our model under different network configurations through simulations. The results demonstrate the advantages of BC-PFS in O-RAN over non-cooperative scheduling despite the impact of blockchain consensus.

The rest of the paper is organized as follows. Section II establishes the O-RAN model. Section III presents the BC-PFS mechanism. Section IV analyzes the impact of delayed scheduling on BC-PFS performance. Section V considers the impact of consensus failure. Section VI identifies the optimal scheduling interval and investigates the trade-off between delayed scheduling and consensus failure. Section VII evaluates the pooling gain. Section VIII provides simulation results, and Section IX concludes the paper.

II. O-RAN Model

A. Multi-operator O-RAN

We consider a multi-operator O-RAN comprising K heterogeneous networks. Each operator network serves N users, so the total number of users is KN, indexed by $i = 1, ..., KN$. As shown in Fig. 1, O-RAN disaggregates base station functions into three components, namely O-RAN central unit (O-CU), O-RAN distributed unit (O-DU), and O-RAN radio unit (O-RU), which can be managed by different vendors or operators. Specifically, the O-CU performs higher-layer RAN functions, while the O-DU handles lower-layer functions, such as radio resource scheduling. The O-DU connects to the O-RU, which supports radio transmission and reception. Moreover, O-RAN provides the near-real-time RAN intelligent controller (Near-RT RIC) and non-real-time RIC (Non-RT RIC) to enable programmable coordination and longer-term policy management, respectively.

Multiple operators leverage open and programmable RAN infrastructures and share a common frequency band through synchronized time-division multiplexing. The instantaneous rate of user i at time t, denoted as $r_i(t)$, is determined by the channel state of its serving operator at that time. We assume that the instantaneous rates $r_i(t)$ and $r_j(t)$ for any two distinct users $i \neq j$ are mutually independent.

B. User Scheduling in O-RAN

In O-RAN, users access the network through the O-RU and report their scheduling-related information. Based on the channel conditions and transmission demands, the O-DU performs user scheduling and allocates the radio resources through control signaling.

User scheduling follows a time-division framework, where time is divided into uniformly spaced slots. Each slot is defined as one scheduling interval of duration τ. In slot n, the scheduling decision is made at time $t_n = n\tau$ and applied over the interval $[t_n, t_{n+1})$. In each slot, the shared spectrum resource is allocated to only one operator to serve a scheduled user. Whether user i accesses the operator is characterized by a binary indicator $I_i(t_n) \in \{0,1\}$, where $I_i(t_n) = 1$ indicates that user i receives transmission service during slot n. Consequently, the achievable rate for user i during slot n is given by $r_i(t) I_i(t_n)$. Furthermore, the throughput achieved by user i up to time t_n is denoted as $\mu_i(t_n)$. We define the long-term average throughput of user i, denoted by $\bar{\mu}_i$, as the limit of the expected instantaneous throughput over time, namely $\bar{\mu}_i = \lim_{n\to\infty} E[\mu_i(t_n)]$.

C. Wireless Channel Model

In Rayleigh fading environment, user i's instantaneous signal-to-noise ratio (SNR), denoted as $\mathrm{SNR}_i(t)$, can be modeled as an exponentially distributed random variable. The statistical characteristics of $\mathrm{SNR}_i(t)$ reflect the stochastic nature of channel quality fluctuations. Based on Shannon's theorem, the instantaneous transmission rate $r_i(t)$ of the user exhibits a logarithmic relationship with $\mathrm{SNR}_i(t)$. Let γ_i denote the mean SNR of the user i. Here, $\bar{r}_i$ characterizes the long-term transmission rate under time-varying channel conditions, while σ_i^2 captures its channel fluctuations.

We use the channel coherence time T_c to denote the duration over which the channel state remains highly correlated. We approximate channel state as unchanged over a duration T_c and varying thereafter. This provides a theoretical foundation for analyzing the coupling relationship between delayed scheduling and channel state, particularly in scenarios where blockchain scheduling intervals mismatch with channel coherence time.

III. Blockchain-based Proportional Fair Scheduling

In this section, we present the BC-PFS to enable efficient and trustworthy user scheduling across operators in multi-operator O-RAN [xx]. BC-PFS executes the PFS rule through smart contracts and records scheduling states and allocation results on-chain, which enables verifiable scheduling execution and traceable resource allocation across operators. These capabilities enable the BC-PFS to support trustworthy collaborative user scheduling among multiple stakeholders in O-RAN.

As shown in Fig. 1, the BC-PFS in multi-operator O-RAN introduces a two-layer blockchain architecture. The edge chain, deployed at the edge O-RAN Cloud (O-Cloud), collects periodically reported user channel state information (CSI), executes the scheduling algorithm via a smart contract, and delivers scheduling decisions to the O-DU to coordinate resource allocation. It also immutably records scheduling states and allocation results, enabling the scheduling process to be traced and verified. Meanwhile, the global chain aggregates information from edge chains and supports overall coordination, cross-operator settlement, and policy optimization.

In principle, BC-PFS in O-RAN enables trustworthy collaborative scheduling across multiple operators. At each scheduling round, only one user is scheduled per time slot according to the proportional fairness:

$$i^*(t_n) = \underset{1\le i\le KN}{\operatorname{argmax}} \frac{r_i(t_n)}{\mu_i(t_n)}. \tag{1}$$

Accordingly, the user throughput is updated iteratively to reflect the latest resource allocations:

$$\mu_i(t_{n+1}) = (1-\alpha)\,\mu_i(t_n) + \alpha r_i(t_n) I_i(t_n), \tag{2}$$

where $\alpha \in (0,1)$ is the smoothing factor that determines the weights of historical throughput and newly achieved rate. Smaller α yields gradual throughput adjustments, while larger α enables faster responses to channel fluctuations. Under the proportional-fair scheduling and throughput-update rules, the long-term average throughput converges to the globally optimal solution of the network utility maximization problem [30]:

$$\text{maximize } U \triangleq \sum_{i=1}^{KN} \ln(\bar{\mu}_i). \tag{3}$$

The workflow of BC-PFS consists of the following four stages: 1) The workflow begins with registration, where users and operators are authenticated through the registration contract, with their verified identities recorded on-chain. 2) After registration, users periodically report their status, including key metrics like CSI and bandwidth requirements, to the O-DU. The O-DU then forwards the corresponding user states to the edge chain. The status reporting contract aggregates and records user rates for subsequent scheduling. 3) During each scheduling interval, the scheduling contract retrieves the latest user rates and historical throughput records. It then executes the PFS rule to determine the scheduled user and update the throughput records. The scheduling decisions are recorded in the new block and then delivered to the corresponding O-DU, which performs the radio resource allocation. 4) Finally, the edge chain aggregates scheduling service records and periodically synchronizes them with the global chain. The settlement contract calculates costs based on operator pricing rules and performs settlement among participants. The global information can be further used to support long-term policy optimization and parameter adjustment by the RIC.

Table I
SYMBOL DEFINITIONS.

Symbols	Explanations
K	Number of operator networks
N	Number of users per operator network
r_i	User rate
μ_i	User throughput
T_c	Channel coherence time
τ	Scheduling interval
T_p	Average blockchain propagation delay
U	Network utility

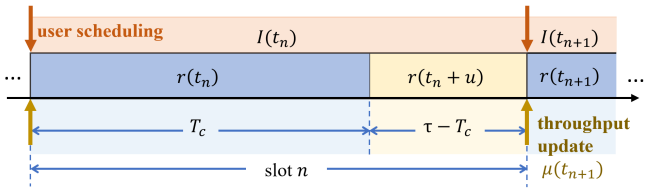


Figure 2. Illustration of delayed scheduled process in the BC-PFS framework.

However, the scheduling interval, determined by the block time, often exceeds the channel coherence time. The scheduling decision made at the beginning of the scheduling interval remains fixed during the interval despite channel variations, resulting in delayed scheduling and suboptimal decisions. Meanwhile, an excessively short scheduling interval may not provide sufficient time for the consensus process to complete, and thus increases the risk of consensus failure. These two effects are coupled through the scheduling interval and jointly affect BC-PFS performance. Therefore, it is necessary to establish an analytical model that captures this mismatch and quantifies its impact on BC-PFS performance in O-RAN.

IV. Delayed Scheduling Model

This section models the BC-PFS under delayed scheduling and analyzes the weak convergence of the user throughput update process. We derive the semi-closed-form expression for user average throughput, together with its approximations and bounds under different limiting conditions. We have summarized important symbols in Table I.

A. Delayed Scheduling

To ensure that the scheduling decisions accurately capture the instantaneous channel state and achieve the network utility maximization objective in (3), the scheduling interval τ is typically set to be smaller than T_c. However, in the BC-PFS, the scheduling interval τ may exceed T_c. This causes the channel state to vary within a scheduling interval and prevents the scheduling decisions from tracking channel dynamics in real-time, ultimately degrading long-term average performance. This mismatch between the fixed scheduling decision and the varying channel state gives rise to delayed scheduling.

We illustrate the delayed scheduling process in Fig. 2. In slot n, the scheduling decision $I(t_n)$ is made at epoch t_n and unchanged in that slot. According to the channel fading model, the user rate r is assumed to be constant for a duration of T_c and fluctuate thereafter due to channel variations. Therefore, the throughput update at t_{n+1} should account for the user rate variations during the slot. Under delayed scheduling, the throughput update for user i should be rewritten as:

$$\mu_i^{\alpha}\left(t_{n+1}\right) = \tag{4}$$
$$\left(1-\alpha\right)\mu_i^{\alpha}\left(t_n\right) + \alpha\frac{1}{\tau}I_i\left(t_n\right)\left(T_c r_i\left(t_n\right) + \int_{T_c}^{\tau} r_i\left(t_n+u\right)du\right),$$

where μ_i^{α} is the sequence related to throughput update with smoothing factor α. The term $T_c r_i\left(t_n\right)$ represents the rate contribution during the coherent time, while $\int_{T_c}^{\tau} r_i\left(t_n+u\right)du$ reflects the channel fading out of the coherent time. The factor $\frac{1}{\tau}$ normalizes the accumulated user rate over the scheduling interval, yielding the average rate during one slot.

B. Weak Convergence Analysis

To characterize the long-term average throughput under delayed scheduling, we first analyze the weak convergence of the stochastic throughput update process in (4) using the stochastic approximation theory. The weak convergence analysis allows the asymptotic behavior of the stochastic process in (4) to be characterized by an associated deterministic ordinary differential equation (ODE), whose limit point captures the long-term average behavior of throughput. To establish this weak convergence, we introduce the following lemma based on the stochastic approximation results in [31]:

Lemma 1. *Under the conditions (C1)-(C7), let θ_i be a discrete-time sequence indexed by $\varepsilon \to 0$, namely*

$$\theta_i^{\varepsilon}\left(t_{n+1}\right) = \theta_i^{\varepsilon}\left(t_n\right) + \varepsilon Y_i^{\varepsilon}\left(t_n\right). \tag{5}$$

For any nondecreasing integer sequence q_ε, there exists a subsequence such that the time-shifted process $\theta_i^{\varepsilon}\left(\varepsilon q_\varepsilon + \cdot\right)$ converges weakly to a continuous-time process $\theta_i\left(t\right)$ satisfying the integral equation:

$$\theta_i\left(t\right) = \theta_i\left(0\right) + \int_0^t \bar{g}_i\left(\boldsymbol{\theta}\left(s\right)\right)ds, \tag{6}$$

where $\boldsymbol{\theta}(\cdot) = \{\theta_i(\cdot),\, i = 1, ..., KN\}$ denotes the continuous-time process with components $\theta_i(\cdot)$, and $\bar{g}_i$ denotes the averaged value of the dynamic process. The conditions (C1)-(C7) are listed as follows:

(C1) $\left\{Y_i^{\varepsilon}\left(t_n\right); \varepsilon, t_n\right\}$ is uniformly integrable.

(C2) Define $\xi^{\varepsilon}\left(t_n\right)$ as the history information up to time t_n. There is a measurable function $g_i^{\varepsilon}\left(\cdot\right)$ and a random variable $\beta^{\varepsilon}\left(t_n\right)$ such that

$$E_t\left[Y_i^{\varepsilon}\left(t_n\right)\right] = g_i^{\varepsilon}\left(\boldsymbol{\theta}^{\varepsilon}\left(t_n\right), \xi^{\varepsilon}\left(t_n\right)\right) + \beta^{\varepsilon}\left(t_n\right), \tag{7}$$

where E_t denotes the expectation conditioned on the history information $\xi^{\varepsilon}\left(t_n\right)$.

(C3) $g_i^{\varepsilon}\left(\boldsymbol{\theta}^{\varepsilon}\left(t_n\right), \xi^{\varepsilon}\left(t_n\right)\right)$ is continuous in $\boldsymbol{\theta}$.

(C4) $\lim_{m,j\to\infty} \frac{1}{m}\sum_{n=j}^{j+m-1} E_t\left[\beta^{\varepsilon}\left(t_n\right)\right] = 0$ in the average sense.

(C5) For $\boldsymbol{\theta}$, the sequence $\left\{g_i^{\varepsilon}\left(\boldsymbol{\theta}^{\varepsilon}\left(t_n\right), \xi^{\varepsilon}\left(t_n\right)\right); \varepsilon, t_n\right\}$ is uniformly integrable.

(C6) For each $\delta > 0$ there is a compact set A_δ such that $\inf_{t,\varepsilon} P\left(\xi^{\varepsilon}\left(t_n\right) \in A_\delta\right) \geq 1 - \delta$.

(C7) For $\boldsymbol{\theta}$, there is a continuous-time function $\bar{g}_i(\cdot)$ such that the following expression holds in probability:

$$\lim_{m,j\to\infty}\frac{1}{m}\sum_{n=j}^{j+m-1} E_t\left[g_i^{\varepsilon}\left(\boldsymbol{\theta}^{\varepsilon}(t_n),\xi^{\varepsilon}(t_n)\right)-\bar{g}_i(\boldsymbol{\theta})\right] \cdot I_{\{\xi^{\varepsilon}(t_n)\in A_{\delta}\}}=0, \tag{8}$$

where $I_{\{\xi^{\varepsilon}(t_n)\in A_{\delta}\}}$ is the indicator function that equals 1 if $\xi^{\varepsilon}(t_n)\in A_{\delta}$ and 0 otherwise.

Now we will show that the stochastic throughput update process in (4) satisfies the conditions (C1)-(C7) in Lemma 1. Define the measurable function $Y_i^{\alpha}(t_n)$ as:

$$Y_i^{\alpha}(t_n)\triangleq \frac{1}{\tau}I_i(t_n)\left(T_c r_i(t_n)+\int_{T_c}^{\tau} r_i(t_n+u)\,du\right)-\mu_i^{\alpha}(t_n). \tag{9}$$

Then, we express the iterative sequence in (4) in the standard stochastic approximation form (5) as:

$$\mu_i^{\alpha}(t_{n+1})=\mu_i^{\alpha}(t_n)+\alpha Y_i^{\alpha}(t_n), \tag{10}$$

where $\alpha\to 0$. First, the user rate $\{r_i(t_n), t_n<\infty\}$ is bounded. Therefore, there exists a constant M such that $\sup_{\alpha,n}|Y_i^{\alpha}(t_n)|\le M$. This implies that $\{Y_i^{\alpha}(t_n);\alpha,t_n\}$ is uniformly integrable, satisfying (C1).

Moreover, based on the wireless channel model established in Section II-C, $\{r_i(t_n), t_n<\infty\}$ constitutes a stationary stochastic process with time-invariant joint distributions. For each bounded random variable $r_i(t_n)$, let $\xi_i(t_n)=\{r_i(\ell):\ell\le t_n\}$ denote the historical rate information up to time slot n, and we define

$$h_i^{\alpha}(\boldsymbol{\mu})\triangleq E_t\left[\frac{1}{\tau}I_i(t_n)\left(T_c r_i(t_n)+\int_{T_c}^{\tau} r_i(t_n+u)\,du\right)\right], \tag{11}$$

where $\boldsymbol{\mu}=\{\mu_i(t_n),\, i=1,...,KN\}$. Therefore, we have

$$E_t[Y_i^{\alpha}(t_n)]=h_i^{\alpha}(\boldsymbol{\mu})-E_t[\mu_i^{\alpha}(t_n)]. \tag{12}$$

In this case, (C2) is satisfied with $\beta^{\alpha}(t_n)=0$. Besides, since $\beta^{\alpha}(t_n)=0$ under all conditions, (C4) is also met. Based on the Bayesian decomposition, (11) can be written as:

$$h_i^{\alpha}(\boldsymbol{\mu})=\frac{1}{\tau}\Pr(I_i(t_n)=1)\left(T_c E_t[r_i(t_n)|I_i(t_n)=1] +E_t\left[\int_{T_c}^{\tau} r_i(t_n+u)\,du|I_i(t_n)=1\right]\right). \tag{13}$$

With the scheduling rule in (1), $h_i^{\alpha}(\boldsymbol{\mu})$ can be expressed as

$$h_i^{\alpha}(\boldsymbol{\mu})=\frac{1}{\tau}\Pr\left(\frac{r_i(t_n)}{\mu_i(t_n)}>\frac{r_j(t_n)}{\mu_j(t_n)},\forall j\ne i\right) \cdot\left(T_c E_t\left[r_i(t_n)|\frac{r_i(t_n)}{\mu_i(t_n)}>\frac{r_j(t_n)}{\mu_j(t_n)},\forall j\ne i\right] +E_t\left[\int_{T_c}^{\tau} r_i(t_n+u)\,du\right]\right). \tag{14}$$

The function $h_i^{\alpha}(\boldsymbol{\mu})$ satisfies local Lipschitz continuity in $\boldsymbol{\mu}\in\mathbb{R}_+^{KN}$. Similarly, owing to the boundedness and stationarity of $\{r_i(t_n), t_n<\infty\}$, $h_i^{\alpha}(\boldsymbol{\mu})$ is continuous in $\boldsymbol{\mu}$ and uniformly integrable. These properties guarantee (C3) and (C5). Meanwhile, according to [30], $\xi_i(t_n)$ can be assumed to reside in a compact sequence space, thereby satisfying (C6).

Leveraging the stationarity of $\{r_i(t_n), t_n<\infty\}$, we further define the steady-state expectation function for user i:

$$\bar{h}_i(\boldsymbol{\mu})=\Pr\left(\frac{r_i}{\mu_i}>\frac{r_j}{\mu_j},\forall j\ne i\right) \cdot\left(\frac{T_c}{\tau}E\left[r_i|\frac{r_i}{\mu_i}>\frac{r_j}{\mu_j},\forall j\ne i\right]+\left(1-\frac{T_c}{\tau}\right)E[r_i]\right), \tag{15}$$

where μ_i and r_i denote the canonical representations of $\mu_i(t_n)$ and $r_i(t_n)$ that exclude time dependencies, respectively. To establish the asymptotic equivalence between $h_i^{\alpha}(\boldsymbol{\mu})$ and $\bar{h}_i(\boldsymbol{\mu})$, we utilize the ergodicity of the stationary process. As t_n is sufficiently large, the dependence of $h_i^{\alpha}(\boldsymbol{\mu})$ on historical data approaches zero. By Khinchine's law of large numbers, for dependent stationary processes that satisfy the asymptotic independence, the average of conditional expectations converges to the steady-state expectation. Therefore, $h_i^{\alpha}(\boldsymbol{\mu})$ converges to $\bar{h}_i(\boldsymbol{\mu})$ in probability:

$$\lim_{m,j\to\infty}\frac{1}{m}\sum_{n=j}^{j+m-1}\left(h_i^{\alpha}(\boldsymbol{\mu})-\bar{h}_i(\boldsymbol{\mu})\right)=0. \tag{16}$$

Consequently, (C7) is satisfied.

In conclusion, we have verified that the throughput update process in (4) satisfies all the conditions (C1)-(C7) in Lemma 1, and thus, it converges weakly to a continuous-time process given in (6). The associated ODE is obtained by taking the time derivative of (6), and its asymptotic behavior is characterized in the following theorem:

Theorem 1. *For BC-PFS under delayed scheduling, as $n\to\infty$, the user throughput $\mu_i(t_n)$ weakly converges to the limit point of the solution of the following ODE system:*

$$\dot{\mu}_i=\bar{h}_i(\boldsymbol{\mu})-\mu_i, i=1,\ldots,KN, \tag{17}$$

where $\bar{h}_i(\boldsymbol{\mu})$ is the steady-state expected rate contribution of user i under event $\frac{r_i}{\mu_i}>\frac{r_j}{\mu_j},\forall j\ne i$, with expression given in (15)*. The limit point of* (17)*, denoted by $\bar{\boldsymbol{\mu}}$, is unique and independent of initial conditions.*

Theorem 1 shows that, under delayed scheduling, the stochastic throughput update process of BC-PFS can still be characterized by a deterministic ODE for analysis. Specifically, the throughput update process accounts for variations in the user rate throughout the scheduling interval, while preserving the stochastic approximation structure. Therefore, user throughput $\mu_i(t_n)$ still weakly converges to the unique asymptotically stable limit point of ODE $\bar{\mu}_i$ as $n\to\infty$. This provides a crucial mathematical foundation for the performance analysis of BC-PFS.

C. Throughput Analysis

The weak convergence result in Theorem 1 establishes that the long-term behavior of BC-PFS is characterized by the limit point of corresponding ODE. Building on this result, we derive the semi-closed-form expression for the long-term average throughput under delayed scheduling. The resulting throughput characterization is given in the following theorem.

Theorem 2. *Under the Rayleigh fading channel, the long-term average throughput of user i in BC-PFS can be approximated as*

$$\tilde{\mu}_i(\tau;T_c) \triangleq \frac{1}{KN}\cdot \left(\frac{T_c}{\tau}\left(\bar{r}_i+\sigma_i\varphi(KN,\eta_i)\right)+\left(1-\frac{T_c}{\tau}\right)\left(\bar{r}_i\Phi(\eta_i)+\sigma_i\phi(\eta_i)\right)\right), \tag{18}$$

where $\eta_i = \frac{\bar{r}_i}{\sigma_i}$ *and* $\varphi(N,\eta_i) \triangleq N\int_{-\eta_i}^{\infty} x\phi(x)\Phi(x)^{N-1}dx$. *The functions* $\phi(\cdot)$ *and* $\Phi(\cdot)$ *are the probability density function (PDF) and cumulative distribution function (CDF) of the standard normal distribution, respectively.*

Proof. Please see Appendix A. ■

The first term in (18) represents the throughput contribution over the coherent time T_c, during which the scheduling decision still relies on valid channel state information. In contrast, the second term corresponds to the remaining interval $\tau - T_c$, during which the channel state varies while the scheduling decision remains unchanged. The second terms captures the contribution under scheduling decisions based on staled channel information. As the scheduling delay increases, the second term becomes more dominant. When $\tau = T_c$, delayed scheduling reduces to the zero-delay case scheduling with only the first term in (18), which is consistent with the result in [32].

We further simplify the user average throughput in (18) by considering the asymptotic case at high SNR.

Corollary 1. *In the high SNR case, the average throughput of user i under BC-PFS can be approximated as*

$$\bar{\mu}_i(\tau;T_c) = \lim_{\eta_i\to\infty}\tilde{\mu}_i(\tau;T_c) = \frac{1}{KN}\left(\bar{r}_i+\frac{T_c}{\tau}\sigma_i\bar{\varphi}(KN)\right), \tag{19}$$

where $\bar{\varphi}(KN) = KN\int_{-\infty}^{\infty} x\phi(x)\Phi(x)^{KN-1}dx$.

The average throughput with delayed scheduling in the BC-PFS comprises two distinct components: the baseline throughput $\frac{\bar{r}_i}{KN}$ that reflects equal resource allocation among users, and the diversity gain from multiple users and operators. Notably, when $\tau = T_c$, the system achieves maximum scheduling gain equivalent to the collaborative PFS adopting accurate channel information. As τ increases, performance degrades due to delayed scheduling.

V. Consensus Failure

Section IV has analyzed the performance of BC-PFS under delayed scheduling. The results indicate that a shorter scheduling interval τ is preferable for mitigating delayed scheduling. However, if τ becomes excessively small, the consensus process may fail to complete within a scheduling round and consequently degrade the performance of BC-PFS.

A. Probability of Consensus Failure

Introduce the consensus protocol. BC-PFS is implemented over the permissioned blockchain jointly maintained by participating operators. Raft, as a representative permissioned consensus mechanism, is suitable for the BC-PFS. In Raft, the consensus delay mainly consists of two message exchange phases, including the proposal propagation from the leader to followers and the acknowledgment collection from followers [33], [34].

When the actual consensus delay is longer than the predetermined scheduling interval τ, the scheduling round cannot complete successfully. This results in no effective service for users during that round and consequently degrades their average throughput. Since each message propagation delay can be modeled as an exponential random variable [35], we introduce a two-stage Erlang distribution with average propagation delay T_p to characterize consensus delay. In this case, the probability of timeout event, namely the probability of consensus failure, can be expressed as:

$$q_f = \exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right). \tag{20}$$

Consequently, the probability of successful scheduling is given by $1-q_f$. According to (20), both the scheduling interval τ and the propagation delay T_p jointly influence the consensus failure probability q_f. A longer scheduling interval τ gives a longer time for blockchain nodes exchanging messages and establishing consensus states, thereby reducing the probability that consensus fails. In contrast, a larger propagation delay T_p means a lengthy message exchange process among blockchain nodes and increases the consensus failure probability.

B. Impact of Consensus Failure

To quantify the impact of consensus failure on user average throughput, we extend the delayed scheduling model in (4) by considering the potential consensus failure. Specifically, the throughput update requires a successful consensus round by introducing the consensus success probability $1-q_f$. Since a scheduling round contributes effective throughput only when consensus is successfully completed, the throughput update for user i becomes:

$$\mu_i(t_{n+1}) = (1-\alpha)\mu_i(t_n) + \alpha(1-q_f)\frac{1}{\tau}I_i(t_n)\left(T_c r_i(t_n)+\int_{T_c}^{\tau} r_i(t_n+u)\,du\right). \tag{21}$$

One can see that (21) has the same stochastic approximation structure as (4), and the weak convergence property and the uniqueness of throughput limit point remain valid. Based on Theorem 2, the user average throughput under both delayed scheduling and consensus failure:

$$\tilde{\mu}_i^f(\tau;T_c,T_p) = \left(1-\exp\left(-\tfrac{\tau}{T_p}\right)\left(1+\tfrac{\tau}{T_p}\right)\right)\tilde{\mu}_i(\tau;T_c). \tag{22}$$

Meanwhile, according to Corollary 1, under high SNR conditions, the user average throughput can be expressed as

$$\bar{\mu}_i^f(\tau;T_c,T_p) = \left(1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)\right)\frac{1}{KN}\left(\bar{r}_i+\frac{T_c}{\tau}\sigma_i\bar{\varphi}(KN)\right), \tag{23}$$

which demonstrates that average throughput is reduced due to consensus failure. The term $1-\exp\left(-\frac{\tau}{T_p}\right)\left(1+\frac{\tau}{T_p}\right)$ reflects the impact of consensus failure, and meanwhile $\frac{T_c}{\tau}$ captures

the throughput loss caused by delayed scheduling. Therefore, the scheduling interval τ establishes a trade-off between delayed scheduling and consensus failure. Reducing τ mitigates delayed scheduling but increases the risk of consensus failure, which should thus be properly selected.

VI. Scheduling Delay versus Consensus Failure

Building upon the analytical model of BC-PFS in Section V, this section investigates the optimal scheduling interval that balances the trade-off between delayed scheduling and consensus failure to maximize the sum throughput in the BC-PFS. Furthermore, we will show how the channel coherence time T_c and blockchain propagation delay T_p influence the system performance and the optimal scheduling interval.

A. Scheduling Interval Optimization

In this subsection, we optimize the scheduling interval τ to maximize the sum of user average throughput in (23), which considers both delayed scheduling and consensus failure. Specifically, the optimization problem can be formulated as

$$\underset{\tau}{\text{maximize}} \quad \sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right). \tag{24}$$

To solve this optimization problem, we first analyze the properties of the objective function $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ with respect to τ. The following theorem establishes the existence and uniqueness of the solution that maximizes $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$.

Theorem 3. *The objective function in* (24) *admits a unique maximizer $\tau^\star$, given by*

$$\tau^\star = \arg_\tau \rho\left(\tau\right) = 0, \tag{25}$$

where $\rho\left(\tau\right)$ has a unique root and can be expressed as $\rho\left(\tau\right) = 1 - \exp\left(\frac{\tau}{T_p}\right) + \frac{\tau}{T_p} + \left(\frac{\tau}{T_p}\right)^2 + \frac{1}{\bar{\varphi}(KN)} \frac{\sum_{i=1}^{KN} \bar{r}_i}{\sum_{i=1}^{KN} \sigma_i} \frac{\tau^3}{T_c T_p^2}$.

Proof. Please see Appendix B. ∎

Theorem 3 proves that the optimization problem in (24) admits a unique optimal solution $\tau^\star$, which is the stationary point of $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$. Moreover, the trend that $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ first increases and then decreases with τ reveals the trade-off between delayed scheduling and consensus failure. Specifically, increasing the scheduling interval τ has two opposite effects: it aggravates delayed scheduling by causing the scheduling decisions to rely more on outdated channel state information, while reducing the consensus failure probability. Over the range $(0, \tau^\star)$, as τ increases, the performance gain from mitigating consensus failure outweighs the loss caused by increased delayed scheduling, and thus the sum of user average throughput $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ increases with τ. At $\tau = \tau^\star$, these two effects are balanced, and $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ reaches its maximum. Conversely, in the interval $(\tau^\star, +\infty)$, further increasing τ yields only limited additional reduction in consensus failure probability as it gradually approaches zero, whereas the throughput loss caused by delayed scheduling becomes more significant. Therefore, $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ transitions to a monotonically decreasing function of τ. Moreover, when $T_p \ll \tau$, the consensus failure probability is negligible, and thus $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ is mainly affected by delayed scheduling and decreases monotonically with τ. The above analysis identifies the theoretical optimum $\tau^\star$. Since our delayed scheduling model assumes $\tau \geq T_c$, the feasibility of $\tau^\star$ should be further examined. If $\tau^\star \geq T_c$, then $\sum_{i=1}^{KN} \bar{\mu}_i^f \left(\tau; T_c, T_p\right)$ attains its maximum at $\tau = \tau^\star$. Otherwise, the maximum is attained at the boundary $\tau = T_c$.

B. Impact of Blockchain Consensus Process

To better understand how T_c and T_p influence the optimal scheduling interval $\tau^\star$, we derive the following theorem.

Proposition 1. *The optimal scheduling interval $\tau^\star$ is strictly decreasing in the channel coherence time T_c and strictly increasing in the blockchain propagation delay T_p.*

Proof. Please see Appendix C. ∎

Proposition 1 characterizes the monotonic relationship between the optimal scheduling interval $\tau^\star$ and the parameters T_c and T_p. First, as T_c decreases, the channel varies faster, and the scheduling decision is outdated for a larger portion of the scheduling interval τ. In this case, reducing τ for more timely scheduling provides relatively limited throughput improvement, whereas a larger τ can effectively mitigate consensus failure. The impact of consensus failure becomes more significant in determining the optimal scheduling interval $\tau^\star$, leading to an increase in $\tau^\star$ as T_c decreases. Second, as T_p decreases, the consensus process can be completed within a shorter interval, which weakens the necessity of increasing τ to avoid consensus failure. The impact of consensus failure on the throughput optimization is less significant, while the delayed scheduling becomes the main factor. Therefore, the optimal scheduling interval $\tau^\star$ shifts toward a smaller value to mitigate the throughput degradation caused by delayed scheduling.

These results reveal that changes in T_c and T_p modify the trade-off between delayed scheduling and consensus failure, thereby shifting the optimal scheduling interval $\tau^\star$. Faster channel variation, i.e., smaller T_c, makes shortening τ to reduce scheduling delay less effective in improving the average throughput compared with enlarging τ to mitigate consensus failure. Meanwhile, a larger blockchain propagation delay T_p increases the necessity of a larger τ. This analysis provides insights into the variation of $\tau^\star$ under different network conditions and helps understand the trade-off between delayed scheduling and consensus failure in BC-PFS for O-RAN.

VII. Pooling Effect with Delay

In this section, we systematically evaluate the pooling effect in O-RAN empowered by BC-PFS. We will derive the conditions for achieving positive pooling gain by balancing the benefits of blockchain-enhanced collaboration against the impact of the consensus process.

For the BC-PFS, multiple operators are jointly scheduled over the shared resources, serving KN users. The network utility of BC-PFS in multi-operator O-RAN is given by

$$\tilde{U}_{\text{coop}} = \sum_{i=1}^{KN} \ln\left(\tilde{\mu}_i^f\left(\tau; T_c, T_p\right)\right). \tag{26}$$

In the non-cooperative PFS scenario where operators schedule independently, there are K isolated networks, each serving N users. The operators schedule independently with real-time channel tracking, i.e., $\tau = T_c$, and the scheduling process is unaffected by consensus failure. Based on (18), the average user throughput in these isolated networks is given by $\frac{1}{N}\left(\bar{r}_i + \sigma_i \varphi(N, \eta_i)\right)$. However, the effective spectrum availability per user is reduced by a factor of K due to time-division frequency sharing across networks, thus the throughput for each user scales as $\frac{1}{K}$. Therefore, the network utility of non-cooperative PFS is given by

$$\tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{1}{KN}\left(\bar{r}_i + \sigma_i \varphi(N, \eta_i)\right)\right). \tag{27}$$

The pooling gain of O-RAN via BC-PFS is defined as the gap between (26) and (27). Although pooling resources across operators can improve network utility, delayed scheduling and consensus failure may diminish this benefit. We therefore investigate whether BC-PFS can achieve positive pooling gain and identify the conditions under which it does. The following theorem gives the conditions on the scheduling interval to guarantee the positive pooling gain.

Theorem 4. *To guarantee the positive pooling gain under delayed scheduling, the scheduling interval τ is required to satisfy the following condition:*

$$\tau < \frac{\varphi\left(KN, 0\right)}{\varphi\left(N, 0\right)} T_c. \tag{28}$$

Proof. Please see Appendix D. ∎

Theorem 4 identifies a conservative range of τ that guarantees positive pooling gain when only delayed scheduling is considered. The upper bound quantifies the maximum scheduling interval that can be tolerated while preserving the pooling gain. As τ increases, the loss caused by outdated scheduling decisions becomes more significant and gradually offsets the benefit of cross-operator resource pooling.

The above theorem only considered the delayed scheduling, and thus preferred a shorter scheduling interval. However, if the consensus failure is considered, the positive pooling gain may require some certain conditions on the channel coherence time T_c and blockchain propagation delay T_p, which is characterized by the following theorem.

Theorem 5. *Under both delayed scheduling and consensus failure, a sufficient condition for the existence of a feasible scheduling interval that yields a positive pooling gain is*

$$\frac{T_p}{T_c} < 1 - \frac{\eta_{\max} + \varphi\left(N, \eta_{\max}\right)}{\eta_{\max} + \varphi\left(KN, \eta_{\max}\right)}, \tag{29}$$

where $\eta_{\max} = \max_{i=1,\ldots,KN} \eta_i$.

Proof. Please see Appendix E. ∎

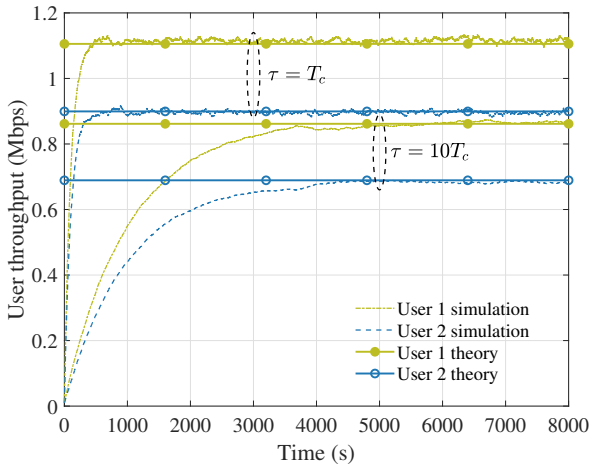


Figure 3. Simulation and theoretical results of user throughput with delayed scheduling.

Theorem 5 shows that T_p needs to be sufficiently small relative to T_c to limit the impact of consensus failure. When (29) is satisfied, there exists a feasible set of τ to achieve positive pooling gain. This result further implies that the consensus process should operate sufficiently faster than the channel variation, especially in scenarios with a small channel coherence time.

VIII. Simulation

In this section, we present our simulation scenarios. We adopt stationary Rayleigh fading with constant and white external noise. The smoothing factor of throughput update is set to $\alpha = 10^{-4}$, which is small enough for the BC-PFS and an acceptable measure of throughput. For all scenarios, we assume the bandwidth of 1 MHz.

To validate the analysis of BC-PFS with delayed scheduling presented in Section IV-C, we simulate a network with two users, each having average SNR values of 4.8 dB and 3 dB, respectively. We assume the channel coherence time $T_c = 10$ ms and examine two scheduling intervals, $\tau = T_c$ and $\tau = 10T_c$, to represent different levels of delayed scheduling. Fig. 3 presents the simulation outcomes and analytic results of Theorem 2 for both users. It can be observed that for a given α, the simulated throughput update consistently demonstrates stable convergence behavior across varying conditions. The convergence values of the simulation demonstrate a strong alignment with the asymptotic analytic results. Additionally, as $\frac{\tau}{T_c}$ increases, the negative impact of delayed scheduling information becomes more evident, which reduces user throughput.

Fig. 4 presents the throughput performance of a user under different SNR conditions in a network with a total of 2 users. It compares the high-SNR theoretical results from Corollary 1, alongside the standard theoretical values from Theorem 2 and corresponding simulation results. The percentage values annotated in the figures represent the relative errors between the high-SNR theoretical values and their corresponding standard theoretical values under each SNR condition. The results demonstrate that the high-SNR theoretical approximation has broad practical applicability. Although this approximation

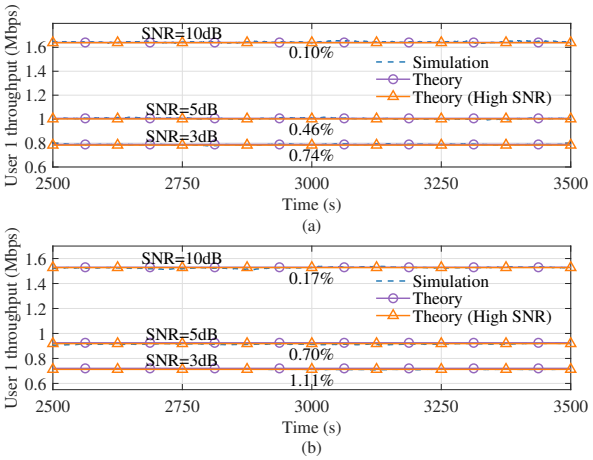


Figure 4. Simulation, theoretical results, and theoretical approximation of user throughput. (a) $\tau = 2T_c$. (b) $\tau = 5T_c$.

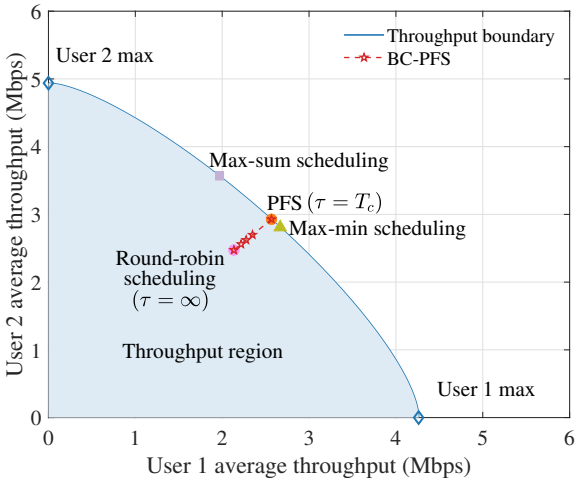


Figure 5. Throughput region: case of two users.

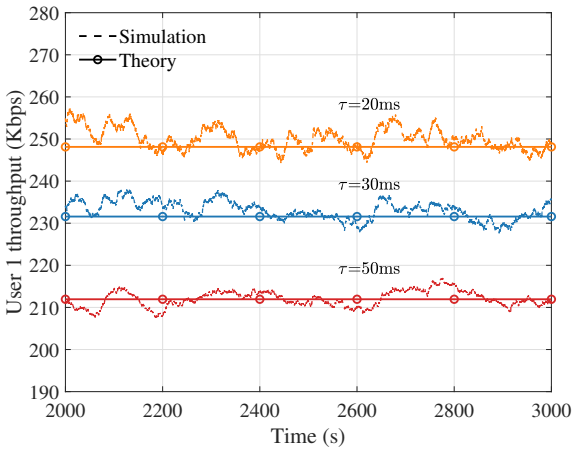


Figure 6. Simulation and theoretical results of user throughput with delayed scheduling and consensus failure.

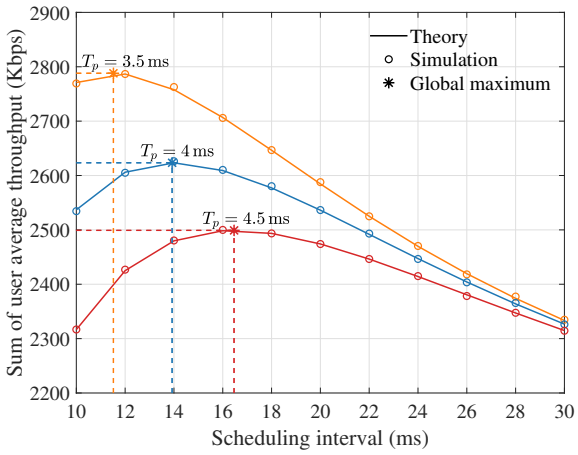
should strictly hold only under the condition that SNR tends to infinity, simulation results reveal that even at moderate SNR levels of 3 dB, the relative error remains below 1.2%. As the SNR increases, the relative error decreases further. This phenomenon indicates that the high-SNR approximation possesses significant practical value in system design, and its applicability can reasonably be extended to moderate SNR regimes. Moreover, while increased delayed scheduling does amplify the discrepancy between theoretical values, its impact remains relatively limited. Comparing the cases of $\tau = 2T_c$ in Fig. 4(a) and $\tau = 5T_c$ in Fig. 4(b), it can be observed that under the same SNR conditions, the increase in delay contributes to an error growth of no more than 0.4%.

Fig. 5 presents the throughput region of a network with two users and the corresponding individual throughput achieved under different scheduling strategies. The dynamic scheduling is performed in each time slot, with only one user scheduled per slot. The figure highlights two extreme points that represent the average throughput each user would achieve if continuously scheduled over the entire duration, reflecting the maximum achievable throughput for each user. Given the Rayleigh fading channel settings, the maximum throughput corresponds to the mean user rate.

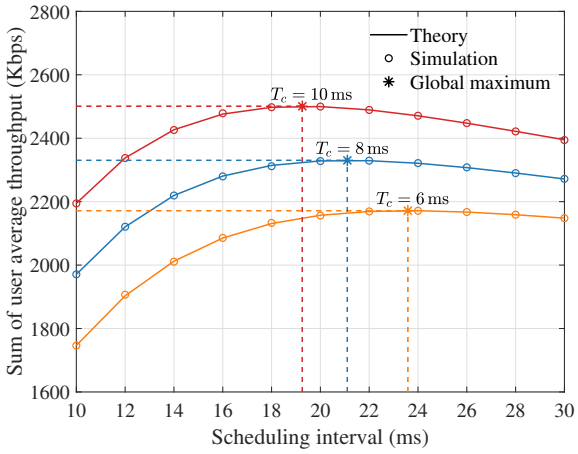
The Maximum-sum scheduling, PFS, and max-min scheduling all operate on the throughput boundary, making these strategies Pareto optimal. That is, for any user, it is impossible to increase the throughput of one user without reducing the throughput of others [36]. Max-sum scheduling seeks to maximize the total network throughput by dynamically allocating resources based on each user's instantaneous rate, which tends to favor users with better channel conditions. In contrast, max-min scheduling prioritizes ensuring the user with the poorest channel conditions receives the most favorable scheduling, thereby maximizing the minimum throughput across all users. The PFS achieves an effective balance between fairness and overall system throughput. When $\tau = T_c$, the BC-PFS is equivalent to PFS. However, as τ increases, the individual throughput under the BC-PFS continuously decreases. When $\tau \to \infty$, the throughput collapses to the level of round-robin scheduling, which allocates resources equally among users regardless of their channel states.

Furthermore, we present simulation results to validate the theoretical analysis of BC-PFS with both delayed scheduling and consensus failure in Section V. Fig. 6 illustrates the user throughput performance in a network of ten users, with the channel coherence time $T_c = 10$ ms and the blockchain propagation delay $T_p = 4$ ms. The simulated throughput closely matches the theoretical results across all scheduling intervals, which confirms the robustness of the analytical framework.

This subsection evaluates the optimal scheduling interval under the joint impact of delayed scheduling and consensus failure. Fig. 7 illustrates the average throughput performance of the network with $K = 2$, $N = 10$, where the numerically determined optimal scheduling interval for maximum throughput is explicitly identified. Fig. 7(a) examines the average throughput of one user under $T_c = 10$ ms for T_p =3.5, 4, and 4.5 ms. The results show that the average throughput first increases and then decreases with longer scheduling intervals, reflecting the trade-off between wireless channel variations and blockchain propagation delay, which aligns with the theoretical



(a)



(b)

Figure 7. Illustration of the optimal scheduling interval. (a) Under different T_p. (b) Under different T_c.

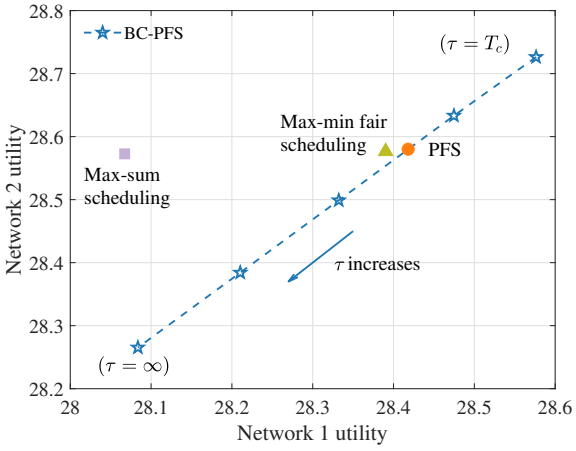


Figure 8. Utility region: case of two networks.

analysis in Theorem 3. Moreover, as T_p increases, the optimal scheduling interval that maximizes throughput shifts toward larger values, consistent with Theorem 1. Concurrently, the average throughput declines due to the higher consensus failure probability induced by longer T_p, which exacerbates throughput loss.

Fig. 7(b) investigates the case where $T_p = 5$ ms under varying $T_c = 6$, 8, and 10 ms. A similar trend of initial increase followed by a decrease in throughput is observed. Furthermore, larger T_c reduces the optimal scheduling interval for peak throughput, again in agreement with Theorem 1. The average throughput improves simultaneously because a longer T_c increases the proportion of synchronized scheduling with accurate channel information. These findings provide valuable insights for parameter optimization in practical BC-PFS systems.

This subsection examines the pooling effect by comparing the network utility under different scheduling methods and network settings. Fig. 8 presents the utility region of two operator networks, each serving two users with $T_c = 10$ ms and $T_p = 2$ ms. The results confirm that, when operators schedule independently within their respective networks, non-cooperative PFS yields superior utility for individual networks compared to max-sum and max-min fair strategies. This is due to the inherent proportional fairness properties of PFS that maximize logarithmic utility functions. Moreover, when the two networks are aggregated under the BC-PFS, the unified system achieves substantially higher overall utility, provided the delayed scheduling remains within appropriate bounds. This demonstrates that, by pooling resources and coordinating scheduling across multiple operators, the BC-PFS can enhance network performance beyond the capabilities of independent scheduling within each isolated network.

Moreover, Fig. 9 presents a comparison of the network utility between the conventional non-cooperative PFS and the BC-PFS. The line represents the network utility of non-cooperative PFS, while the bars show the utility gain of BC-PFS over non-cooperative PFS under different scheduling intervals. Fig. 9(a) is evaluated under varying numbers of operator networks K with each maintaining sixty users. The BC-PFS corresponds to the network utility of the aggregated network, while non-cooperative PFS represents the sum of the utilities of multiple independent networks. The results indicate that for $\tau = T_c$, $\tau = 1.1T_c$, and $\tau = 1.2T_c$, the network utility under the BC-PFS is significantly higher than that under non-cooperative PFS. Moreover, as the number of operator networks grows, the advantage of the BC-PFS becomes increasingly pronounced. This demonstrates that the aggregation of multiple operator networks enhances the overall network utility, particularly as the network scale expands.

Fig. 9(b) maintains twenty operator networks with varying user density N. It is obvious that the BC-PFS consistently outperforms non-cooperative PFS under all considered acceptable delayed-scheduling settings, which further validates the benefits of cooperative resource pooling. Consistent with the trend observed in Fig. 9(a), the network utility degrades as $\frac{\tau}{T_c}$ increases, reflecting the performance penalty caused by higher delayed scheduling. Meanwhile, when the user density grows, both scheduling methods exhibit improved network utility due to multi-user diversity gain, while the performance gap between them, namely the pooling gain, also widens. This underscores the performance advantages of the BC-PFS in

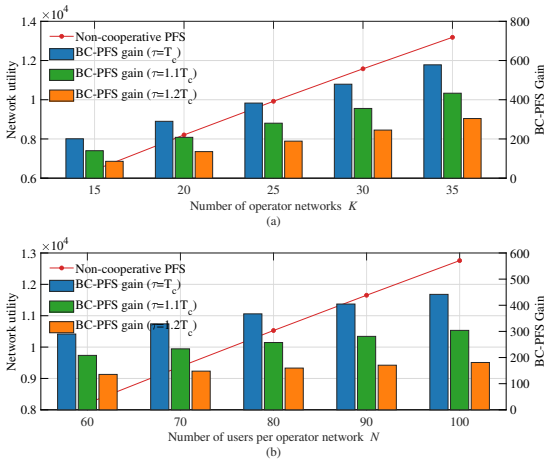


Figure 9. Pooling gain of BC-PFS over non-cooperative PFS. (a) Under different K. (b) Under different N.

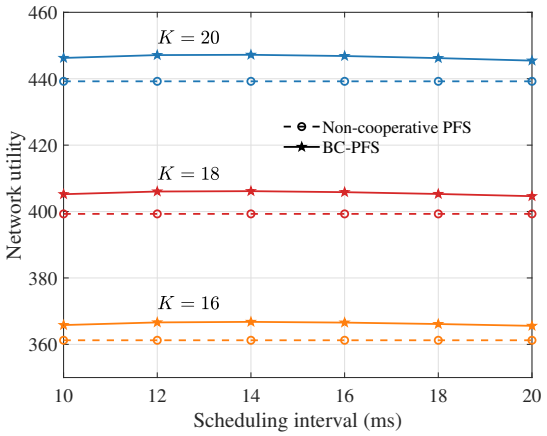


Figure 10. Comparison of network utility between non-cooperative PFS and BC-PFS under varying τ.

densely populated networks.

Furthermore, we evaluate the pooling effect of BC-PFS with delayed scheduling and consensus failure. Fig. 10 examines the relationship between network utility and scheduling intervals under the network configuration with $T_p = 4$ ms, $T_c = 10$ ms, and $N = 2$. The results demonstrate that the BC-PFS consistently outperforms non-cooperative PFS across different operator counts and scheduling intervals, validating the fundamental advantage of resource pooling among multiple operators. Notably, the network utility of the BC-PFS follows a similar non-monotonic pattern with respect to τ as the average throughput. As τ increases, the network utility first increases and then declines. The optimal performance occurs when the trade-off between wireless channel variations and blockchain propagation delays is balanced. These insights verify the theoretical analysis in Section VII and highlight the importance of carefully selecting scheduling intervals to enhance the performance of blockchain-enhanced O-RAN.

IX. Conclusion

In this paper, we analyze the impact of the blockchain consensus process on the BC-PFS for O-RAN. We first develop an analytical model for BC-PFS considering delayed scheduling and prove the weak convergence of the user throughput update process. Based on the weak convergence property, we derive the semi-closed-form expressions of the long-term average throughput and its approximation under high SNR or large scheduling interval. Consensus failures are further incorporated into the BC-PFS analysis framework. We uncover the trade-off between the delayed scheduling and consensus failure in the BC-PFS and examine the optimal scheduling interval that balances these two factors. Furthermore, we quantify the pooling effect and provide insights for the system parameter optimization. Simulation results support our analysis and illustrate the impact of various parameters on average throughput and network utility. This work provides crucial insights for system parameter configuration in practical implementations.

Appendix

A. Proof of Theorem 2

Proof. Taking steady-state expectations on both sides of (4) yields:

$$\bar{\mu}_i = \frac{T_c}{\tau} \lim_{n\to\infty} E\left[r_i\left(t_n\right) I_i\left(t_n\right)\right] + \frac{1}{\tau} \lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n+u\right) du I_i\left(t_n\right)\right]. \tag{30}$$

We define c_i and d_i as scheduling metrics reflecting current and outdated channel states, respectively, given by

$$c_i \triangleq \lim_{n\to\infty} E\left[r_i\left(t_n\right) I_i\left(t_n\right)\right], \tag{31}$$

$$d_i \triangleq \lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n+u\right) du I_i\left(t_n\right)\right]. \tag{32}$$

We will first analyze the component c_i. Based on the Bayesian decomposition and scheduling rule in (1), we derive:

$$\begin{aligned} c_i &= \lim_{n\to\infty} E\left[r_i\left(t_n\right)|I_i\left(t_n\right)=1\right] \Pr\left(I_i\left(t_n\right)=1\right) \\ &= \lim_{n\to\infty} \int_0^{\infty} x f_{r_i}(x) \Pr\left(\frac{x}{\mu_i\left(t_n\right)} > \frac{r_j\left(t_n\right)}{\mu_j\left(t_n\right)}, \forall j\neq i\right) dx, \end{aligned} \tag{33}$$

where $f_r(x)$ represents the PDF of user rate. From Theorem 1, $\mu_i(t_n)$ converges to $\bar{\mu}_i$ as $t_n \to \infty$. Therefore, taking the limit, we obtain:

$$\begin{aligned} c_i &= \int_0^{\infty} x f_{r_i}(x) \Pr\left(\frac{r_j \bar{\mu}_i}{\bar{\mu}_j} < x, \forall j\neq i\right) dx \\ &= \int_0^{\infty} x f_{r_i}(x) \prod_{\forall j\neq i} F_{r_j}\left(\frac{\bar{\mu}_j}{\bar{\mu}_i} x\right) dx, \end{aligned} \tag{34}$$

where $F_r(x)$ represents the CDF of the user rate.

In Rayleigh fading environment, given $\bar{r}_i \le \bar{r}_j$ for two users i, j , we have $\frac{\sigma_j}{\sigma_i} \le \frac{\bar{\mu}_j}{\bar{\mu}_i} \le \frac{\bar{r}_j}{\bar{r}_i}$ under the proportional fairness definition, where $\bar{\mu}_i$ and $\bar{\mu}_j$ are the average throughput of

users i, j, respectively [32]. Therefore, we have the following expressions through the variable substitution $y = \frac{x-\bar{r}_i}{\sigma_i}$:

$$c_i \leq \sigma_i \int_{-\eta_i}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \cdot \prod_{\forall \bar{r}_i \leq \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\bar{r}_j}{\bar{r}_i}x\right) \prod_{\forall \bar{r}_i > \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\sigma_j}{\sigma_i}x\right) dy, \quad (35)$$

$$c_i \geq \sigma_i \int_{-\eta_i}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \cdot \prod_{\forall \bar{r}_i \leq \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\sigma_j}{\sigma_i}x\right) \prod_{\forall \bar{r}_i > \bar{r}_j \wedge j \neq i} F_{r_j}\left(\frac{\bar{r}_j}{\bar{r}_i}x\right) dy, \quad (36)$$

Then, c_i can be approximated by the following expression that lies between above bounds:

$$c_i \approx \sigma_i \int_{-\eta_i}^{\infty} \left(y\sigma_i + \bar{r}_i\right) f_{r_i}\left(y\sigma_i + \bar{r}_i\right) \prod_{\forall j \neq i} F_{r_j}\left(y\sigma_i + \bar{r}_i\right) dy. \quad (37)$$

In fact, the user rate over Rayleigh fading channels can be accurately modeled by a normal distribution [32], [37], i.e., approximately $r_i \sim \mathcal{N}\left(\bar{r}_i, \sigma_i^2\right)$. For normally distributed r_i, its PDF and CDF can be written as:

$$f_{r_i}(x) = \frac{1}{\sigma_i}\phi\left(\frac{x-\bar{r}_i}{\sigma_i}\right), \; F_{r_i}(x) = \Phi\left(\frac{x-\bar{r}_i}{\sigma_i}\right), \quad (38)$$

Substituting (38) into (37) yields:

$$c_i \approx \quad (39)$$
$$\sigma_i \int_{-\frac{\bar{r}_i}{\sigma_i}}^{\infty} y\phi(y)\,\Phi(y)^{KN-1}\,dy + \frac{\bar{r}_i}{KN}\left(1 - \Phi\left(-\frac{\bar{r}_i}{\sigma_i}\right)^{KN}\right).$$

Since $\Phi(-\eta_i)^{KN} \ll 1$, (39) is approximated by

$$c_i \approx \sigma_i \int_{-\eta_i}^{\infty} y\phi(y)\,\Phi(y)^{KN-1}\,dy + \frac{\bar{r}_i}{KN} \triangleq \tilde{c}_i. \quad (40)$$

Meanwhile, the delayed component of the average throughput, namely d_i, can be written as:

$$d_i = \quad (41)$$
$$\lim_{n\to\infty} E\left[\int_{T_c}^{\tau} r_i\left(t_n + u\right) du | I_i\left(t_n\right) = 1\right] \Pr\left(I_i\left(t_n\right) = 1\right).$$

With the similar analysis for c_i, we have:

$$\begin{aligned} d_i &= \left(\tau - T_c\right) \int_0^{\infty} x f_{r_i}(x)\,dx \int_0^{\infty} f_{r_i}(x) \prod_{\forall j \neq i} F_{r_j}\left(\frac{\bar{\mu}_j}{\bar{\mu}_i}x\right) dx \\ &= \frac{\tau - T_c}{KN}\left(\sigma_i\phi(\eta_i) + \bar{r}_i\Phi(\eta_i)\right)\left(1 - \Phi(-\eta_i)^{KN}\right) \\ &\approx \frac{\tau - T_c}{KN}\left(\sigma_i\phi(\eta_i) + \bar{r}_i\Phi(\eta_i)\right) \triangleq \tilde{d}_i. \end{aligned} \quad (42)$$

According to (30), the approximation of average throughput $\tilde{\mu}_i(\tau; T_c)$ can be written as:

$$\tilde{\mu}_i(\tau; T_c) = \frac{T_c}{\tau}\tilde{c}_i + \frac{1}{\tau}\tilde{d}_i. \quad (43)$$

Notably, combining (40) and (42), we can obtain that (43) is equivalent to (18), which completes the proof. ■

B. Proof of Theorem 3

Proof. The first-order derivative of $\sum_{i=1}^{KN} \bar{\mu}_i^f(\tau; T_c, T_p)$ with respect to τ is

$$\begin{aligned} \frac{\partial}{\partial\tau}\left(\sum_{i=1}^{KN} \bar{\mu}_i^f(\tau; T_c, T_p)\right) &= \frac{\exp\left(-\frac{\tau}{T_p}\right)\sum_{i=1}^{KN}\bar{r}_i}{\tau^2 KNA} \\ &\cdot \left(1 - \exp\left(\frac{\tau}{T_p}\right) + \frac{\tau}{T_p} + \left(\frac{\tau}{T_p}\right)^2 + \frac{A\tau^3}{T_p^2}\right) \\ &\triangleq \frac{\exp\left(-\frac{\tau}{T_p}\right)\sum_{i=1}^{KN}\bar{r}_i}{\tau^2 KNA}\rho(\tau), \end{aligned} \quad (44)$$

where $A = \frac{\sum_{i=1}^{KN}\bar{r}_i}{T_c\bar{\varphi}(KN)\sum_{i=1}^{KN}\sigma_i}$. To analyze the stationary points of $\sum_{i=1}^{KN} \bar{\mu}_i^f(\tau; T_c, T_p)$, we examine the behavior of $\rho(\tau)$. The derivatives of $\rho(\tau)$ are given by

$$\begin{aligned} \rho'(\tau) &= \frac{1}{T_p}\left(1 - \exp\left(\frac{\tau}{T_p}\right) + \frac{2\tau}{T_p} + \frac{3A\tau^2}{T_p}\right), \quad (45) \\ \rho''(\tau) &= \frac{1}{T_p^2}\left(2 - \exp\left(\frac{\tau}{T_p}\right) + 6A\tau\right), \\ \rho'''(\tau) &= \frac{1}{T_p^3}\left(-\exp\left(\frac{\tau}{T_p}\right) + 6AT_p\right), \\ \rho''''(\tau) &= -\frac{1}{T_p^4}\exp\left(\frac{\tau}{T_p}\right) < 0. \end{aligned}$$

Hence, $\rho'''(\tau)$ is strictly decreasing and $\lim_{\tau\to+\infty}\rho'''(\tau) = -\infty$. If $\lim_{\tau\to 0^+}\rho'''(\tau) > 0$, $\rho''(\tau)$ first increases then decreases with τ. Otherwise, $\rho''(\tau)$ monotonically decreases with τ. In either case, we have $\lim_{\tau\to 0^+}\rho''(\tau) > 0$ and $\lim_{\tau\to+\infty}\rho''(\tau) = -\infty$. Therefore, $\rho''(\tau) = 0$ has a unique root, denoted as τ_1. Then, $\rho'(\tau)$ increases monotonically on $(0, \tau_1]$ and decreases monotonically on $[\tau_1, +\infty)$. Similarly, $\rho'(\tau) = 0$ also has a unique root, and $\rho(\tau)$ first increases and then decreases. Furthermore, $\rho(\tau) = 0$ has a unique root $\tau^\star$, with $\rho(\tau) > 0$ for $\tau < \tau^\star$ and $\rho(\tau) < 0$ for $\tau > \tau^\star$.

Since $\frac{\exp\left(-\frac{\tau}{T_p}\right)\sum_{i=1}^{KN}\bar{r}_i}{\tau^2 KNA} > 0$, the sign of $\frac{\partial}{\partial\tau}\left(\sum_{i=1}^{KN}\bar{\mu}_i^f(\tau; T_c, T_p)\right) = 0$ is determined by $\rho(\tau)$. Therefore, $\sum_{i=1}^{KN}\bar{\mu}_i^f(\tau; T_c, T_p)$ increases on $(0, \tau^\star]$ and decreases on $[\tau^\star, +\infty)$, attaining its unique global maximum at $\tau = \tau^\star$. ■

C. Proof of Proposition 1

Proof. From the proof of Theorem 3, the stationary point $\tau^\star$ satisfies $\rho(\tau^\star) = 0$. As $\tau^\star$ implicitly depends on both T_p and T_c, we may compute the partial derivatives with respect to these parameters to analyze this dependence. The partial derivatives yield:

$$\frac{\partial\tau^\star}{\partial T_p} = \frac{\tau^\star}{T_p D}\left(D - \sum_{i=1}^{KN} A\tau^{\star 2}\right), \quad \frac{\partial\tau^\star}{\partial T_c} = \frac{1}{D}\sum_{i=1}^{KN}\frac{A\tau^{\star 3}}{T_c},$$

where

$$D = \sum_{i=1}^{KN}\left(\left(T_p\left(1 - \exp\left(\frac{\tau^\star}{T_p}\right)\right) + 2\tau^\star\right)\sigma_i + 3A\tau^{\star 2}\right).$$

As demonstrated in the proof of Theorem 3, we have $\rho'(\tau^\star) < \rho'(\tau_0) = 0$. Expressing $\rho'(\tau^\star)$ using (45) gives $\rho'(\tau^\star) = \frac{D}{T_p^2} < 0$, confirming that $D < 0$. Therefore, $\frac{\partial \tau^\star}{\partial T_p} > 0$ and $\frac{\partial \tau^\star}{\partial T_c} < 0$, which proves that the optimal scheduling interval $\tau^\star$ monotonically decreases with T_c and increases with T_p. ■

D. Proof of Theorem 4

Proof. A sufficient condition for positive pooling gain under delayed scheduling is that each logarithmic term in

$$\tilde{U}_{\text{coop}} - \tilde{U}_{\text{iso}} = \sum_{i=1}^{KN} \ln\left(\frac{KN\tilde{\mu}_i^f(\tau; T_c, T_p)}{\bar{r}_i + \sigma_i \varphi(N, \eta_i)}\right). \tag{46}$$

is positive with $q_f = 0$, i.e.,

$$\frac{T_c}{\tau}(\eta_i + \varphi(KN, \eta_i)) + \left(1 - \frac{T_c}{\tau}\right)(\eta_i \Phi(\eta_i) + \phi(\eta_i)) > \eta_i + \varphi(N, \eta_i). \tag{47}$$

Since $\phi(\eta_i) - \eta_i(1 - \Phi(\eta_i)) = \int_{\eta_i}^{\infty}(x - \eta_i)\phi(x)dx > 0$, we have

$$\eta_i \Phi(\eta_i) + \phi(\eta_i) > \eta_i. \tag{48}$$

Hence, based on (47), a stronger sufficient condition is

$$\eta_i + \frac{T_c}{\tau}\varphi(KN, \eta_i) > \eta_i + \varphi(N, \eta_i),$$

equivalently $\tau < \frac{\varphi(KN,\eta_i)}{\varphi(N,\eta_i)} T_c$. Since $\frac{\partial}{\partial \eta_i}\left(\frac{\varphi(KN,\eta_i)}{\varphi(N,\eta_i)}\right) > 0$, we have $\frac{\varphi(KN,\eta_i)}{\varphi(N,\eta_i)} > \frac{\varphi(KN,0)}{\varphi(N,0)}$. Therefore, a more explicit sufficient condition is

$$\tau < \frac{\varphi(KN, 0)}{\varphi(N, 0)} T_c. \tag{49}$$

■

E. Proof of Theorem 5

Proof. A sufficient condition for positive pooling gain under delayed scheduling and consensus failure is that each logarithmic term in (46) is positive. Combining with (48), a stronger sufficient condition is

$$\left(\eta_i + \frac{T_c}{\tau}\varphi(KN, \eta_i)\right)\left(1 - \exp\left(-\frac{\tau}{T_p}\right)\left(1 + \frac{\tau}{T_p}\right)\right) - (\eta_i + \varphi(N, \eta_i)) > 0. \tag{50}$$

Since $f(x) = x\exp(-x)(1 + x) \leq f(\frac{1+\sqrt{5}}{2}) < 1$, we have

$$1 - \exp(-x)(1 + x) > 1 - \frac{1}{x}.$$

Then a further sufficient condition for (50) is

$$\left(\eta_i + \frac{T_c}{\tau}\varphi(KN, \eta_i)\right)\left(1 - \frac{T_p}{\tau}\right) - (\eta_i + \varphi(N, \eta_i)) > 0. \tag{51}$$

We substitute $\tau = T_c$ into (51) gives

$$\frac{T_p}{T_c} < 1 - \frac{\eta_i + \varphi(N, \eta_i)}{\eta_i + \varphi(KN, \eta_i)}. \tag{52}$$

A further sufficient condition for (52) is

$$\frac{T_p}{T_c} < 1 - \frac{\eta_{\max} + \varphi(N, \eta_{\max})}{\eta_{\max} + \varphi(KN, \eta_{\max})}, \tag{53}$$

where $\eta_{\max} = \max_{i=1,\ldots,KN} \eta_i$. This constitutes a sufficient condition on T_c and T_p for the existence of positive pooling gain. ■

REFERENCES

[1] M. Polese, M. Dohler, F. Dressler, M. Erol-Kantarci, R. Jana, R. Knopp, and T. Melodia, "Empowering the 6g cellular architecture with open ran," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 2, pp. 245–262, Feb. 2024.

[2] B. Agarwal, R. Irmer, D. Lister, and G.-M. Muntean, "Open RAN for 6g networks: Architecture, use cases and open issues," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 2881–2924, Apr. 2025.

[3] O-RAN Alliance, "Spectrum sharing based on shared O-RUs," O-RAN Next Generation Research Group Research Report, Oct. 2023. [Online]. Available: https://mediastorage.o-ran.org/ngrg-rr/nGRG-RR-2023-05-Spectrum_Sharing_with_Shared_O-RU-v1_0.pdf

[4] S. Acharya, S. Li, N. Jiang, Y. Wu, W. Xie, W. Lou, and Y. T. Hou, "Real-time scheduling for GAA users' coexistence in CBRS using O-RAN architecture," *IEEE Trans. Cognit. Commun. Networking*, vol. 12, pp. 2573–2589, Jul. 2026.

[5] O-RAN Alliance, "O-RAN use cases and deployment scenarios: Towards open and smart RAN," O-RAN Alliance, White Paper, Feb. 2020, accessed: Jul. 27, 2026. [Online]. Available: https://www.o-ran.org/o-ran-resources/o-ran-use-cases-and-deployment-scenarios-whitepaper-february-2020

[6] Y. Chen, Y. T. Hou, W. Lou, J. H. Reed, and S. Kompella, "O-M3: Real-time multi-cell mimo scheduling in 5G O-RAN," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 2, pp. 339–355, Feb. 2024.

[7] A. E. Matemu, M. Kim, and K. Lee, "Joint scheduling, O-RU association, and power allocation in O-RAN via model-based optimization and PPO-based reinforcement learning," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 20 102–20 117, Jul. 2026.

[8] A. S. Abdalla, P. S. Upadhyaya, V. K. Shah, and V. Marojevic, "Toward next generation open radio access networks: What O-RAN can and cannot do!" *IEEE Network*, vol. 36, no. 6, pp. 206–213, 2022.

[9] J. Akram, A. Anaissi, R. S. Rathore, R. H. Jhaveri, and A. Akram, "Digital twin-driven trust management in open ran-based spatial crowdsourcing drone services," *IEEE Trans. Green Commun. Netw.*, vol. 8, no. 3, pp. 1061–1075, Sep. 2024.

[10] P. Porambage, M. Christopoulou, B. Han, M. A. Habibi, H. Bogucka, and P. Kryszkiewicz, "Security, privacy, and trust for open radio access networks in 6g," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 332–361, Dec. 2024.

[11] X. Ling, Y. Le, S. Chen, J. Wang, and X. Zhou, "Blockchain-enabled decentralized services and networks: Assessing roles and impacts," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 6, pp. 2141–2154, Jun. 2025.

[12] M. Wu, X. Ling, J. Wang, Y. Le, K. Huang, B. Cao, Y. Huang, D. Niyato, Z. Ding, and X. You, "Blockchain-driven resource management in wireless communications and networks: Models, approaches, and applications," *IEEE Commun. Surv. Tutorials*, vol. 28, pp. 2306–2344, Aug. 2026.

[13] L. Giupponi and F. Wilhelmi, "Blockchain-enabled network sharing for O-RAN in 5G and beyond," *IEEE Network*, vol. 36, no. 4, pp. 218–225, Jul. 2022.

[14] G. O. Boateng, D. Ayepah-Mensah, D. M. Doe, A. Mohammed, G. Sun, and G. Liu, "Blockchain-enabled resource trading and deep reinforcement learning-based autonomous RAN slicing in 5G," *IEEE Trans. Netw. Serv. Manage.*, vol. 19, no. 1, pp. 216–227, 2022.

[15] Z. Yu, Z. Chang, T. Mikkonen, V. Frascolla, and S. Mumtaz, "Blockchain cooperative spectrum management for wi-fi and lte-unlicensed coexistence networks," *IEEE Trans. Commun.*, vol. 74, pp. 6409–6425, Mar. 2026.

[16] H. Kim and Y. Han, "A proportional fair scheduling for multicarrier transmission systems," *IEEE Communications Letters*, vol. 9, no. 3, pp. 210–212, 2005.

[17] H. Zhou, P. Fan, and J. Li, "Global proportional fair scheduling for networks with multiple base stations," *IEEE Trans. Veh. Technol.*, vol. 60, no. 4, pp. 1867–1879, Feb. 2011.

[18] T. Faisal, M. Dohler, S. Mangiante, and D. R. Lopez, "BEAT: Blockchain-enabled accountable and transparent network sharing in 6G," *IEEE Commun. Mag.*, vol. 60, no. 4, pp. 52–56, 2022.

[19] H. Xu, Z. Zhou, L. Zhang, Y. Sun, and C.-L. I, "BE-RAN: Blockchain-enabled Open RAN for 6G with DID and privacy-preserving communication," in *2024 IEEE Globecom Workshops (GC Wkshps)*, Cape Town, SA, Dec. 2024, pp. 1015–1021.

[20] H. Zhang, S. Leng, Y. Wei, and J. He, "A blockchain enhanced coexistence of heterogeneous networks on unlicensed spectrum," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7613–7624, Apr. 2022.

[21] L. Qian, C. Liu, and J. Zhao, "User connection and resource allocation optimization in blockchain empowered metaverse over 6G wireless communications," *IEEE Trans. Wireless Commun.*, vol. 24, no. 1, pp. 19–34, 2025.

[22] H. Zhang, S. Leng, and H. Chai, "A blockchain enhanced dynamic spectrum sharing model based on proof-of-strategy," in *Proc. 2020 IEEE Int. Conf. Commun. (ICC)*, Dublin, IE, Jul. 2020, pp. 1–6.

[23] Kaspa. [Online]. Available: https://kaspa.org/

[24] Fastest blockchains by transactions per second. [Online]. Available: https://chainspect.app/dashboard?gainers=false&order=asc&sort=blocktime

[25] H. Jung, T. â. Kwon, K. Cho, and Y. Choi, "REACT: Rate adaptation using coherence time in 802.11 WLANs," *Comput. Commun.*, vol. 34, no. 11, pp. 1316–1327, Jul. 2011.

[26] N. Ishikawa, R. Rajashekar, C. Xu, M. El-Hajjar, S. Sugiura, L.-L. Yang, and L. Hanzo, "Differential-detection aided large-scale generalized spatial modulation is capable of operating in high-mobility millimeter-wave channels," *IEEE J. Sel. Top. Signal Process.*, vol. 13, no. 6, pp. 1360–1374, Oct. 2019.

[27] W. W.-L. Li, Y. J. Zhang, A. M.-C. So, and M. Z. Win, "Slow adaptive OFDMA systems through chance constrained programming," *IEEE Trans. Signal Process.*, vol. 58, no. 7, pp. 3858–3869, Jul. 2010.

[28] C. Xu, P. Zhang, H. Yu, and Y. Li, "Dynamic blockchain-empowered trustworthy end-edge collaborative computing via rotating multi-agent DRL," *IEEE Trans. Wireless Commun.*, vol. 24, no. 6, pp. 4864–4878, Jun. 2025.

[29] Y. Amoussou-Guenou, A. Del Pozzo, M. Potop-Butucaru, and S. Tucci-Piergiovanni, "Correctness of Tendermint-Core blockchains," in *22nd International Conference on Principles of Distributed Systems (OPODIS'18)*, ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 125, Hong Kong, China, Dec. 2019, pp. 16:1–16:16.

[30] H. Kushner and P. Whiting, "Convergence of proportional-fair sharing algorithms under general conditions," *IEEE Trans. Wireless Commun.*, vol. 3, no. 4, pp. 1250–1259, Jul. 2004.

[31] K. Harold and G. Yin, *Stochastic approximation and recursive algorithm and applications*. New York, NY, USA: Springer, 2003.

[32] E. Liu, Q. Zhang, and K. K. Leung, "Asymptotic analysis of proportionally fair scheduling in Rayleigh fading," *IEEE Trans. Wireless Commun.*, vol. 10, no. 6, pp. 1764–1775, Jun. 2011.

[33] D. Huang, X. Ma, and S. Zhang, "Performance analysis of the Raft consensus algorithm for private blockchains," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 50, no. 1, pp. 172–181, Jan. 2020.

[34] H. Luo, X. Yang, H. Yu, G. Sun, B. Lei, and M. Guizani, "Performance analysis and comparison of nonideal wireless PBFT and RAFT consensus networks in 6g communications," *IEEE Internet Things J.*, vol. 11, no. 6, pp. 9752–9765, Mar. 2024.

[35] E. Sakic and W. Kellerer, "Response time and availability study of RAFT consensus in distributed SDN control plane," *IEEE Trans. Netw. Serv. Manage.*, vol. 15, no. 1, pp. 304–318, Mar. 2018.

[36] R. Etkin, A. Parekh, and D. Tse, "Spectrum sharing for unlicensed bands," *IEEE J. Sel. Areas Commun.*, vol. 25, no. 3, pp. 517–528, Apr. 2007.

[37] P. Smith and M. Shafi, "An approximate capacity distribution for MIMO systems," *IEEE Trans. Commun.*, vol. 52, no. 6, pp. 887–890, 2004.